Student Worksheet

Activity 3E: Keypad Lock

Purpose: Learn how to create a circuit that uses light, sound, and motion to detect moving objects. Use a servo motor to add a moving pop-up to animate your alarm.

Materials:

2 LEDs, two 220 or 330 Ω resistors, 15 jumper wires, servo motor, membrane key pad

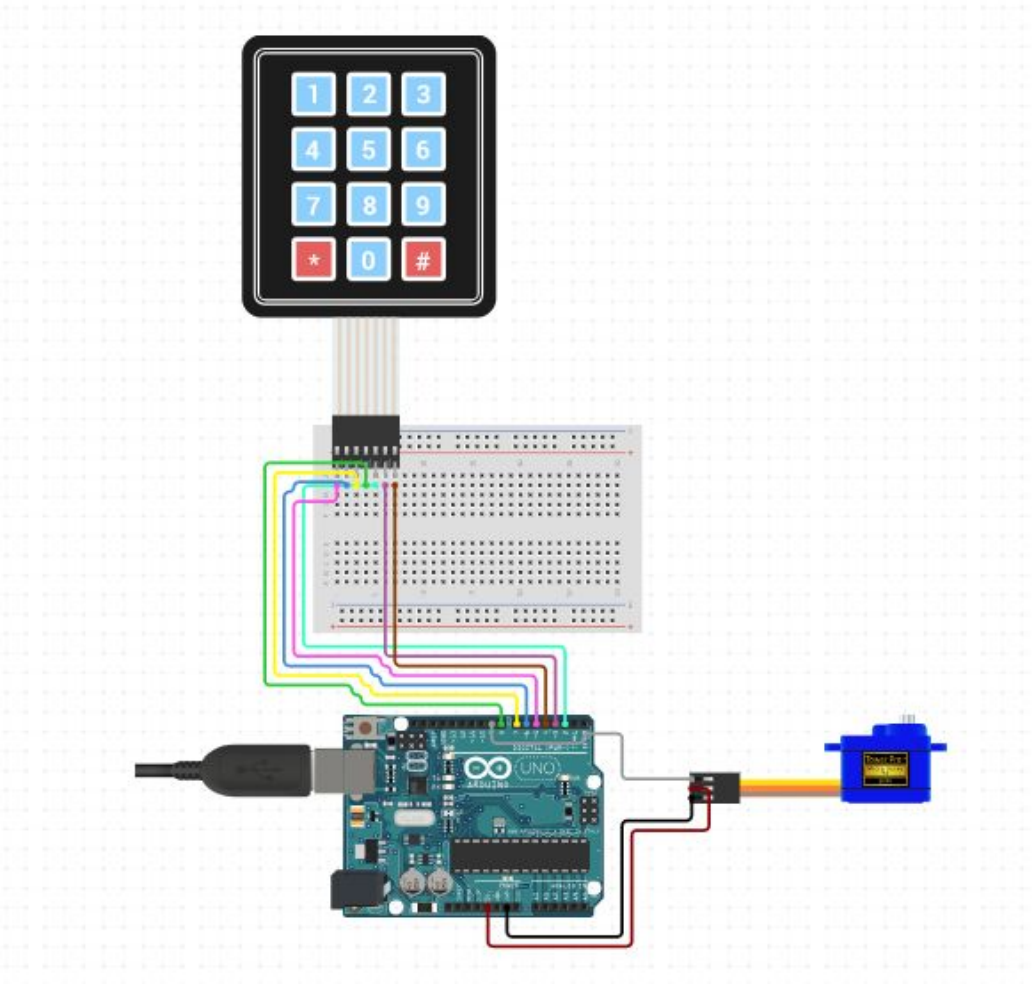


New Components:

Arduino compatible keypad: A keypad is often needed to provide input to an Arduino system, This keypad has 12 buttons, arranged in a telephone-line 3x4 grid. They are quite thin and can easily be mounted wherever they are needed.

Procedure:

1. Wire the project using the following schematic.



2. Copy and paste the below Arduino Code into the [Codebender website](). Hook up your arduino to your computer using the USB cable.

```
#include <Servo.h>
#include <Keypad.h>
Servo ServoMotor;
char* password = "753"; // change the password here, just pick any 3 numbers
int position = 0;
const byte ROWS = 4;
const byte COLS = 3;
char keys[ROWS][COLS] = {
{'1','2','3'},
{'4','5','6'},
```

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```
{'7','8','9'},
{'*','0','#'}
};
byte rowPins[ROWS] = { 8, 7, 6, 5, };
byte colPins[COLS] = {4,  3, 2, };
Keypad keypad = Keypad( makeKeymap(keys), rowPins, colPins, ROWS, COLS );
int RedpinLock = 12;
int GreenpinUnlock = 13;
void setup()
{
pinMode(RedpinLock, OUTPUT);
pinMode(GreenpinUnlock, OUTPUT);
ServoMotor.attach(11);
LockedPosition(true);
}
void loop()
{
char key = keypad.getKey();
if (key == '#')
{
position = 0;
LockedPosition(true);
}
if (key == password[position])
{
position ++;
}
if (position == 3)
{
LockedPosition(false);
}
delay(100);
}
void LockedPosition(int locked)
{
if (locked)
{
digitalWrite(RedpinLock, HIGH);
digitalWrite(GreenpinUnlock, LOW);
ServoMotor.write(11);
}
else
{
digitalWrite(RedpinLock, LOW);
digitalWrite(GreenpinUnlock, HIGH);
ServoMotor.write(90);
}
```

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4. Your program is now loaded.

5. Save your sketch as “3E-Keypad Lock.”

What you should see:

The red LED will be on. When you enter the code “753”, the servo will rotate 90 degrees (in the open position). Your green LED will turn on to indicate that the servo has opened the lock. When you press the “#” key, the servo will rotate back to its original position and the red LED will go back on to indicate that the lock is once again locked.

Challenges:

1. Go into the code and change your passwords. Upload your code to codebender to see if it worked.

2. Add some bling to your project. Add a sound, some LED lights, or even a second servo. Code your project to have a green LED lit when the servo is activated and then a warning sound and a red light lit when the servo is going to return to its original position.

3. Build a “safe” or “lock box” to keep your valuables secure. To gain access to your safe, you need to enter a code. You will then need to lock the box when you are done using a different code..

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